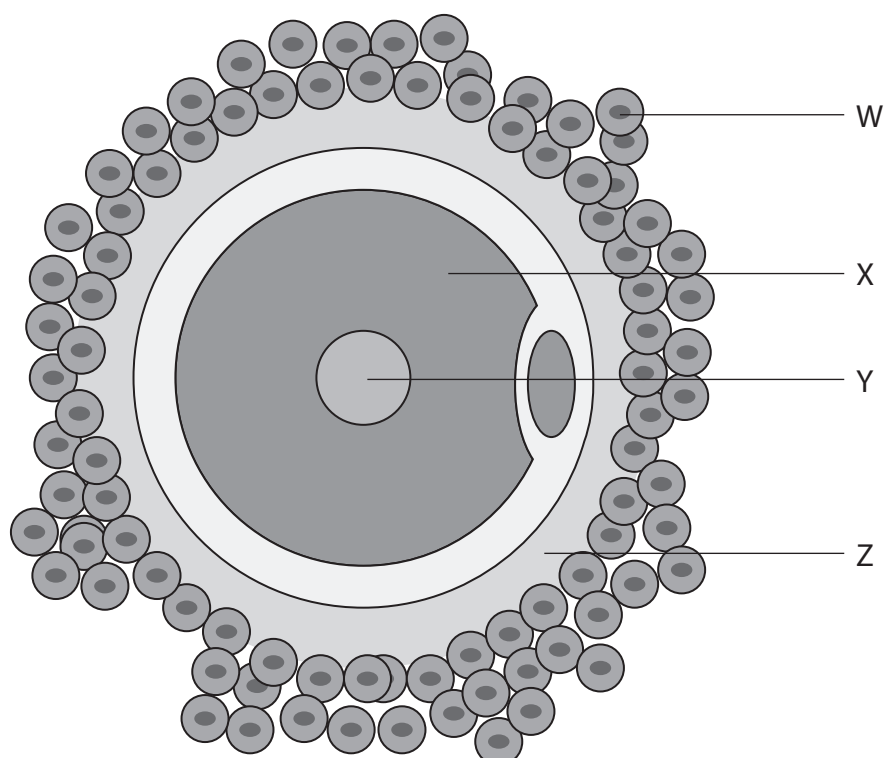


8 Some organisms can produce gametes for sexual reproduction.

- (a) The diagram shows an incomplete drawing of a human egg cell, drawn by a student.



- (i) State the letter which shows the zona pellucida.

(1)

- (ii) The student did not include the structures that contain enzymes involved in hardening the zona pellucida.

Name these structures.

(1)

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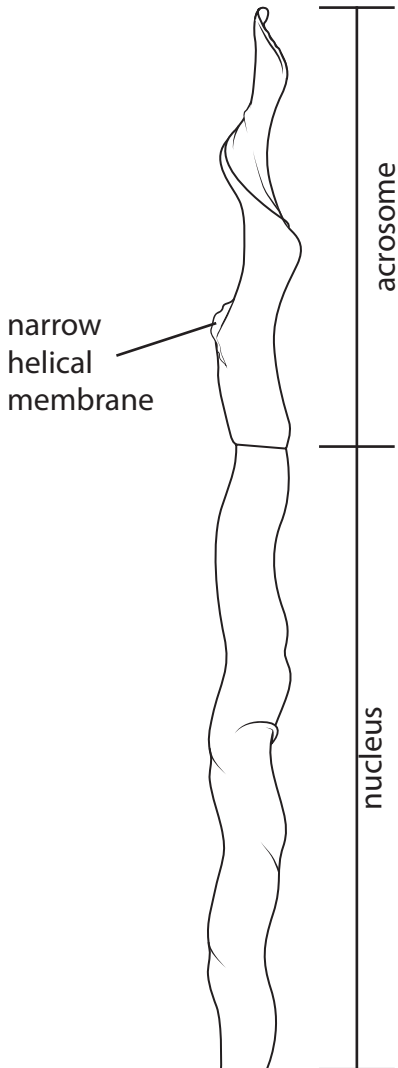
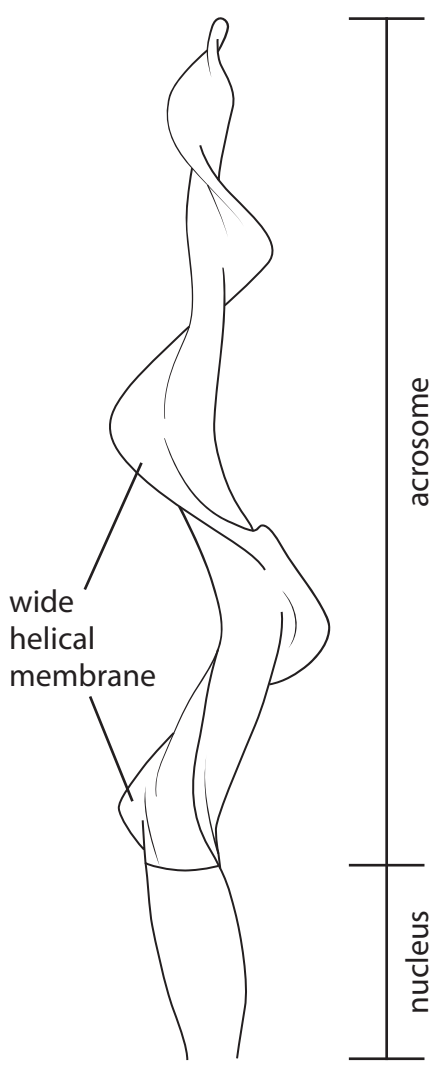
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(b) The table shows some information about two birds.

Both birds have a helical membrane around part of their sperm cells.

Information	Nuthatch	Reed bunting
Photograph	 (Source: Drew Buckley / Alamy Stock Photo)	 (Source: Angela Yates / Alamy Stock Photo)
Reproductive behaviour	Female has one male partner	Female mates with several males in a short period of time.
Drawing of head of sperm cell		

(i) How many of the following statements about sperm cells are correct?

- The mid-piece of the sperm cell contains many mitochondria for anaerobic respiration.
- The acrosome is a membrane bound sac containing digestive enzymes.
- The nucleus contains circular DNA.

(1)

☐ **A** none

☐ **B** one

☐ **C** two

☐ **D** three

(ii) Sperm cells with a larger acrosome to nucleus ratio have an increased swimming speed.

Calculate the acrosome to nucleus ratio for the reed bunting sperm cell shown in the table.

(1)

Answer

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



- (c) An individual organism that can produce both egg cells and sperm cells is said to be a hermaphrodite.

Blue dragons (*Glaucus atlanticus*) are sea slugs that are hermaphrodites and can be found in the Indian Ocean.

The photograph shows a blue dragon.



(Source: BIOSPHOTO / Alamy Stock Photo)

Each individual can produce both egg cells and sperm cells for sexual reproduction with another blue dragon.

Their sperm cells cannot fertilise their own egg cells.

Suggest reasons why it may be advantageous for blue dragons to be hermaphrodites.

(3)



(d) The table shows some information about a population of 100 cats.

Colour of cat			
	Black	Black	White
Genotype	BB	Bb	bb
Part of Hardy–Weinberg equation	p^2	$2pq$	q^2
Number of cats			16

Complete the table to show how many cats would have the homozygous dominant genotype and how many cats would have a heterozygous genotype.

Use the equation

$$p^2 + 2pq + q^2 = 1$$

(3)

(Total for Question 8 = 14 marks)

TOTAL FOR PAPER = 80 MARKS